

THERMODYNAMICS



States of matter

2

Matter – Definition:

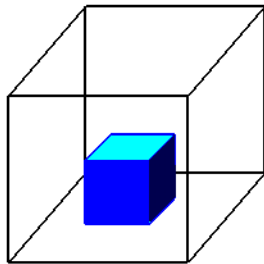
- Aggregation of atoms/ions/molecules which come within the scope of human experience i.e. any substance that has mass and occupies space.

States of Matter:

- Matter exists in one of the following six states due to the energy of its particles: *Gaseous, Liquid, Solid, Plasma, Bose-Einstein* and *Filament*. The states of matter are also known as phases of matter or states of aggregation.
- The division of matter into states is not always simple. How, for example, should chocolate spread be considered ?.

Four states of matter

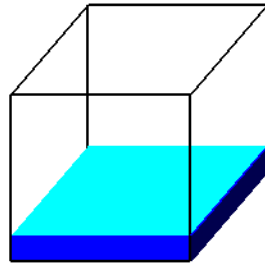
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Solid

Holds Shape

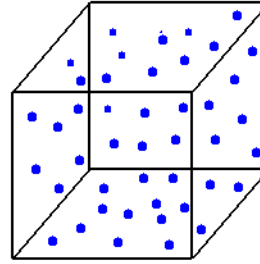
Fixed Volume



Liquid

Shape of Container
Free Surface

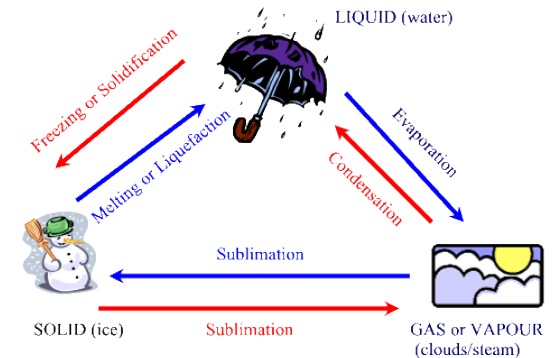
Fixed Volume



Gas

Shape of Container

Volume of Container



Matter is made up of particles which are in continual random motion

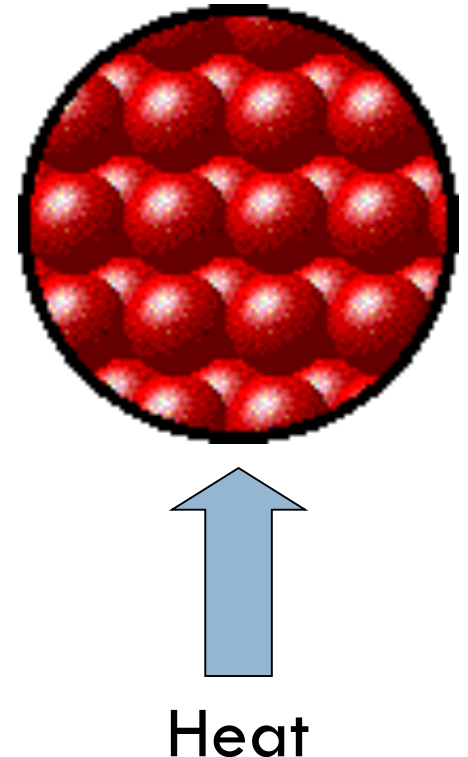
- Based upon particle arrangement
- Based upon energy of particles
- Based upon distance between particles

State	Determinant	Structure
Gas (simplest state of matter)	• Kinetic energy is more important than potential energy. • Intermolecular force are weak.	• Collection of particles in continuous random chaotic motion. • Has no well-defined surface. • No limit on extent of space or volume it can occupy.
Liquid	• Potential energy is more important than kinetic energy. • Intermolecular forces are strong	• No long-range order but vestiges of crystal lattice structure at short range. • Has a surface and place a limit on the extent of space or volume it can occupy.
Solid	• Potential energy is more important than kinetic energy. • Intermolecular forces are strong	Generally possesses crystal lattice structure –well defined shape

States of matter - Solids

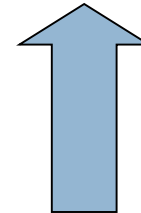
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- Particles of solids are tightly packed, vibrating about a fixed position
- Solids have a definite shape and a definite volume



States of matter - Liquid

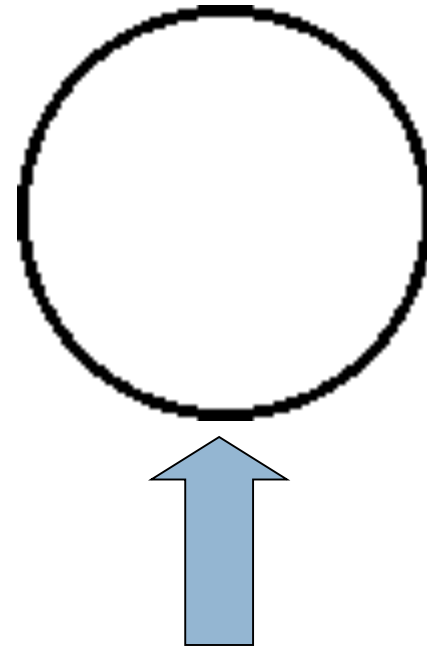
- Particles of liquids are tightly packed, but are far enough apart to slide over one another.
- Liquids have an indefinite shape and a definite volume.



Heat

States of matter - Gas

- Particles of gases are very far apart and move freely
- Gases have an indefinite shape and an indefinite volume



Heat

Phase changes

8

**Description
of Phase
Change**

**Term for Phase
Change**

**Heat Movement During
Phase Change**

**Solid to
liquid**

Melting

**Heat goes into the
solid as it melts.**

**Liquid to
solid**

Freezing

**Heat leaves the liquid
as it freezes.**

Phase changes

9

Description of
Phase Change

Term for Phase
Change

Heat Movement During
Phase Change

Liquid to gas

Vaporization
(boiling and
evaporation)

Heat goes into the liquid
as it vaporizes.

Gas to liquid

Condensation

Heat leaves the gas as it
condenses.

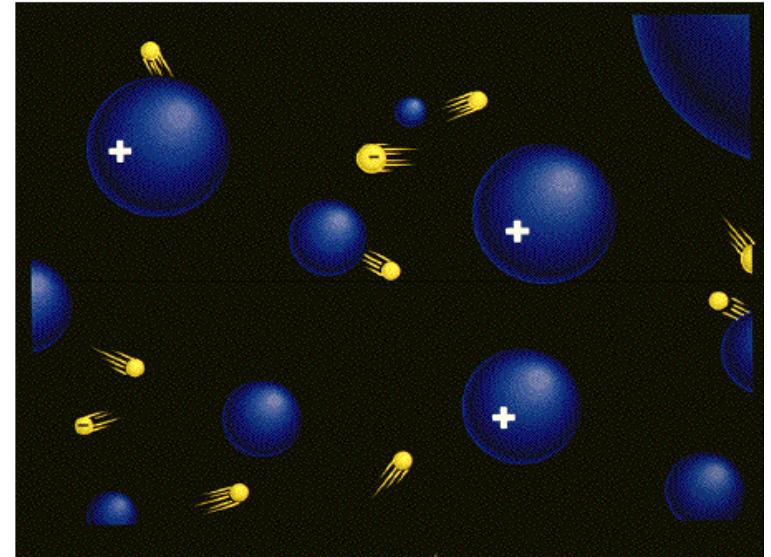
Solid to gas

Sublimation

Heat goes into the solid
as it sublimates.

States of matter - Plasma

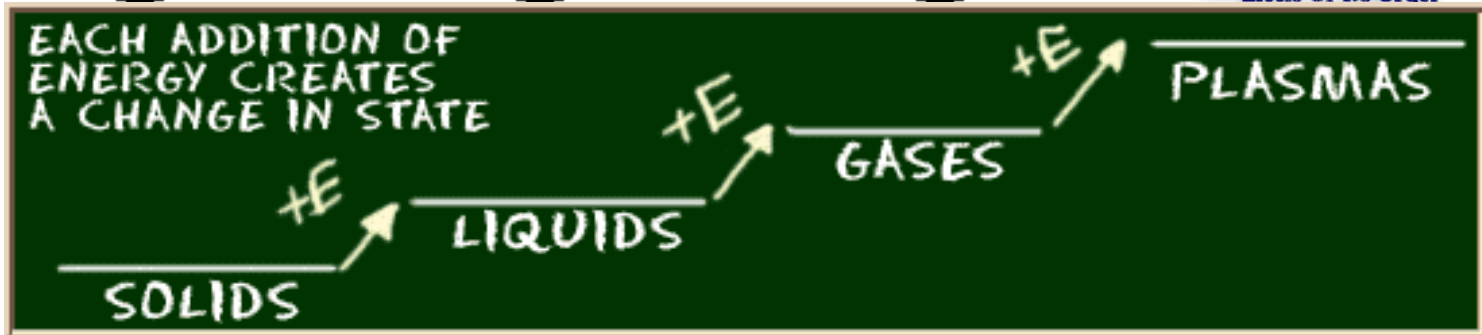
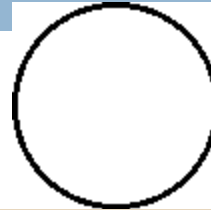
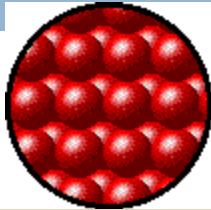
- A plasma is an ionized gas.
- A plasma is a very good conductor of electricity and is affected by magnetic fields.
- Plasmas, like gases, have an indefinite shape and an indefinite volume.



Plasma

States of matter

11



SOLID

Tightly packed, in a regular pattern
Vibrate, but do not move from place to place

LIQUID

Close together with no regular arrangement.
Vibrate, move about, and slide past each other

GAS

Well separated with no regular arrangement.
Vibrate and move freely at high speeds

PLASMA

Has no definite volume or shape and is composed of electrical charged particles

Basic concepts of thermodynamics

12

- The study of the flow of heat or any other form of energy into or out of a system as it undergoes a physical or chemical transformation.
- Thermodynamics deals with the inter-conversion of various Kinds of energy and changes in physical properties that are involved.

Three empirical laws

13

- The study of thermodynamics is based on three broad generalisations derived from well established experimental results
- These generalisations are known as the First, Second and Third law of thermodynamics

Scope of thermodynamics

14

- Most of the important laws of Physical Chemistry can be derived from the laws of thermodynamics.
- It tells whether a particular physical or chemical change can occur under a given set of conditions of temperature, pressure and concentration.
- It also helps in predicting how far a physical or chemical change can proceed, until the equilibrium conditions are established.

Limitations of thermodynamics

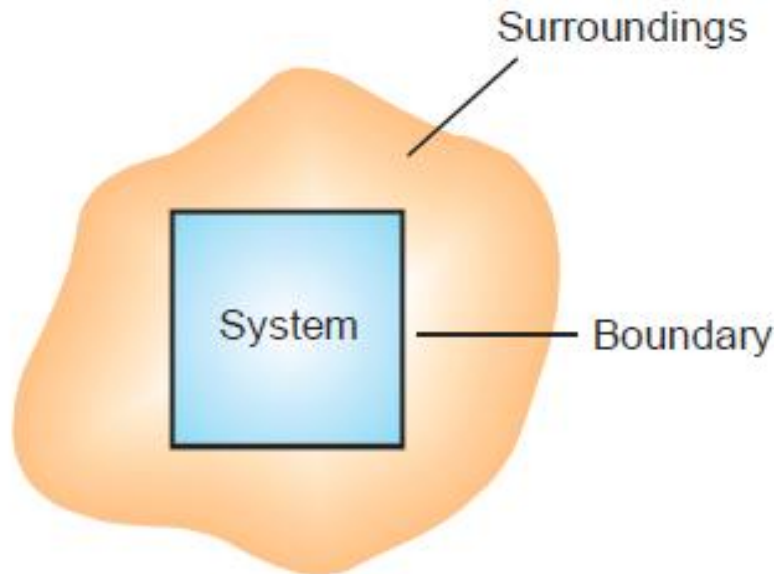
15

- Thermodynamics is applicable to macroscopic systems consisting of matter in bulk and not to microscopic systems of individual atoms or molecules. It ignores the internal structure of atoms and molecules.
- Thermodynamics does not bother about the time factor. That is, it does not tell anything regarding the rate of a physical change or a chemical reaction. It is concerned only with the initial and the final states of the system.

Thermodynamic systems

16

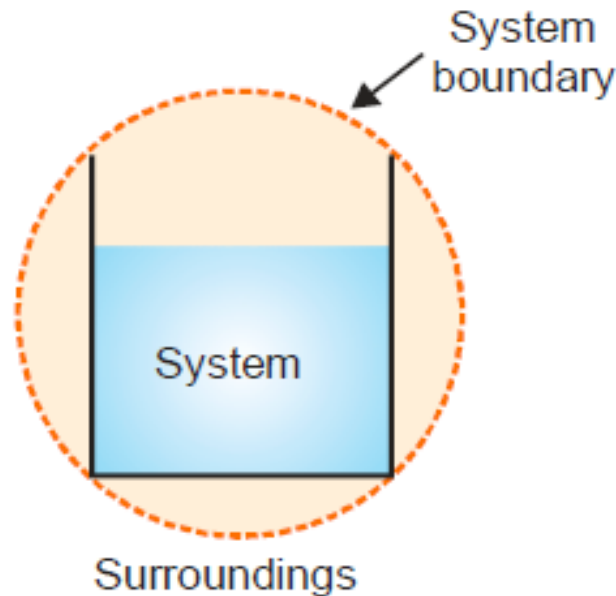
- A part of the physical universe that is separated from the rest of the universe by real or imaginary boundaries under consideration or discussion.



Thermodynamic surroundings

17

- The part of the universe outside the boundary of the system is referred to as surroundings



Homogeneous systems

18

- A homogeneous System is the system that is uniform throughout
- A homogeneous system is made of one phase only
- Examples are : a pure single solid, liquid or gas, mixtures of gases, and true solution of a solid in a liquid
- A phase is a homogeneous, physically distinct and mechanically separable portion of a system

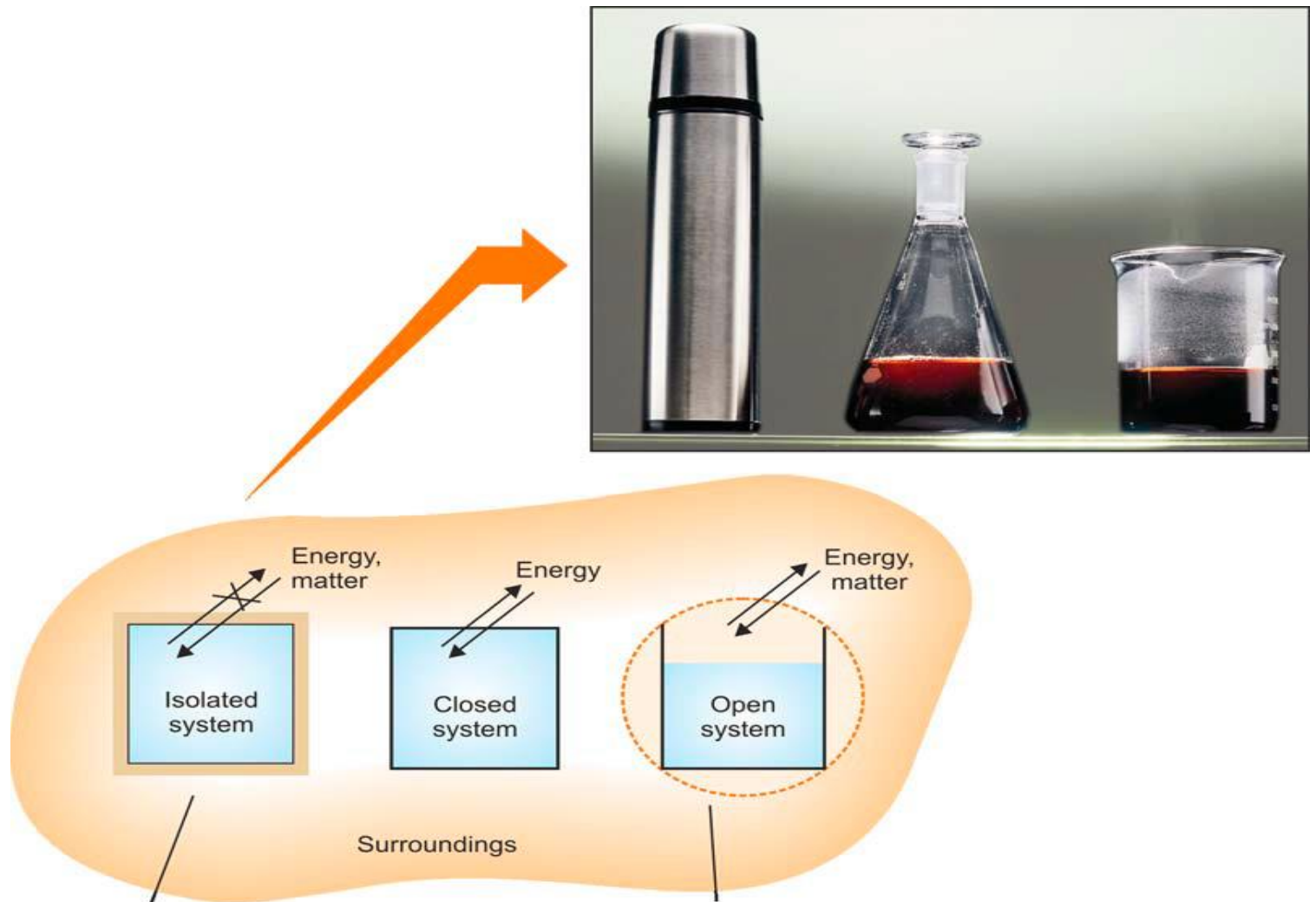
Heterogeneous systems

19

- A heterogeneous system is one which consists of two or more phases. In other words it is not uniform throughout.
- Examples of heterogeneous systems are : ice in contact with water, ice in contact with vapour etc. Here ice, water and vapour constitute separate phases.

Types of thermodynamic systems

20



Isolated system

21

- When the boundary is both sealed and insulated, no interaction is possible with the surroundings.
- Therefore, an isolated system is one that can transfer neither matter nor energy to and from its surroundings.

Closed system

22

- A closed system is one which cannot transfer matter but can transfer energy in the form of heat, work and radiation to and from its surroundings.
- The boundary of a closed system is sealed but not insulated.

Open system

23

- Therefore, an open system is one which can transfer both energy and matter to and from its surroundings.
- The boundary of an open system is open and un-insulated.

Intensive and extensive properties

24

- The macroscopic or bulk properties of a system (volume, pressure, mass, etc.) can be divided into two classes:
 - Intensive properties
 - Extensive properties

Intensive properties

25

- A property which does not depend on the quantity of matter present in the system.
- Examples of intensive properties are pressure, temperature, density, and concentration.

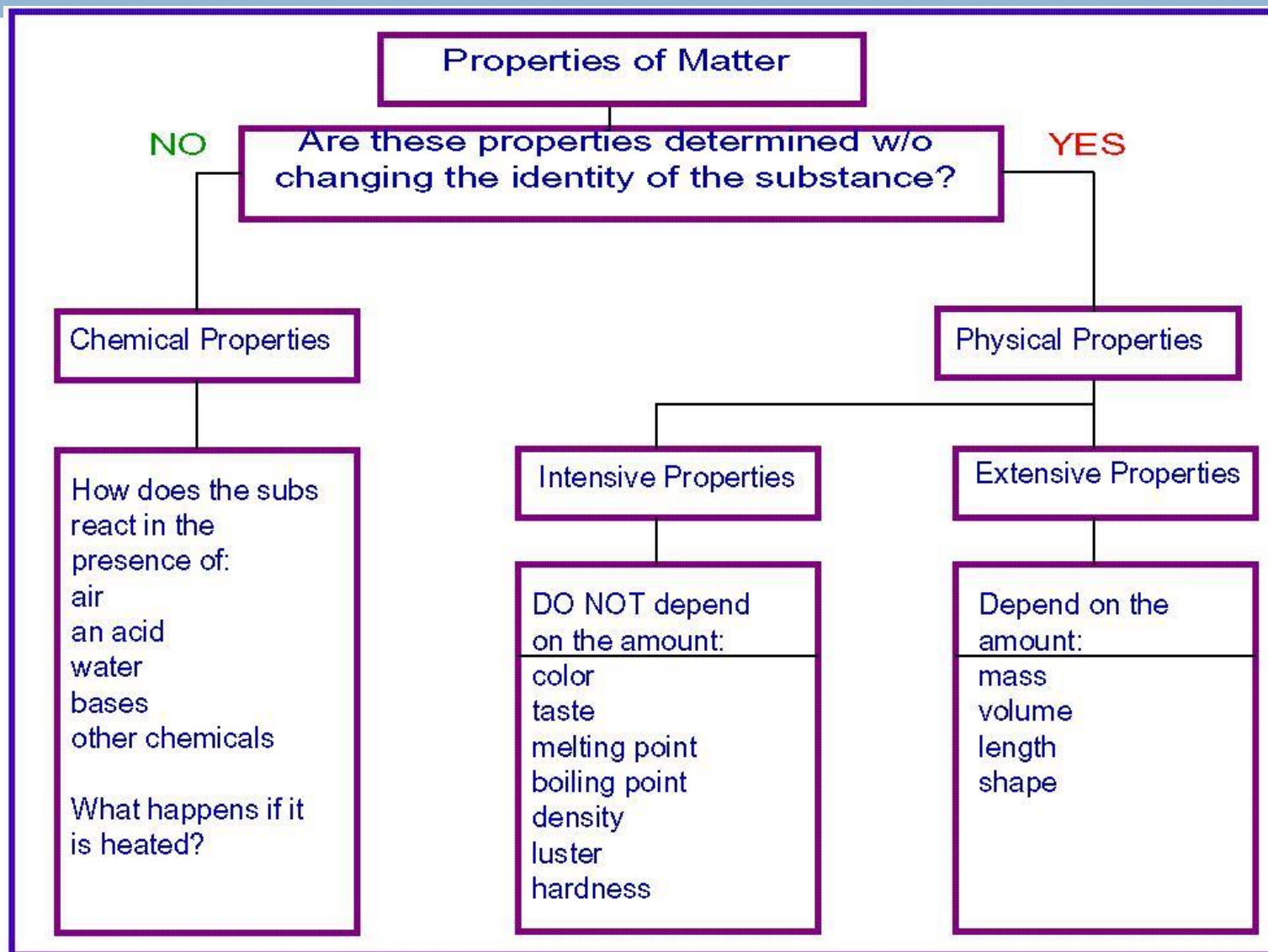
Extensive properties

26

- A property that does depend on the quantity of matter present in the system.
- Some examples of extensive properties are volume, number of moles, enthalpy, entropy, and Gibbs' free energy.

Intensive and extensive properties

27



Intensive and extensive properties

28

Intensive properties		Extensive properties
Temperature	Surface tension	Mass
Pressure	Refractive index	Volume
Density	Viscosity	Internal energy
Boiling point	Freezing point	Enthalpy, Entropy

State of a system

29

- A thermodynamic system is said to be in a certain state when all its properties are **fixed**.
- The fundamental properties which determine the state of a system are **pressure** (P), **temperature** (T), **volume** (V), **mass** and **composition**.
- Since a change in the magnitude of such properties alters the state of the system, these are referred to as **State variables** or **State functions** or **Thermodynamic parameters**.

State of a system

30

- A change of system from the initial state to the final state (2nd state) will be accompanied by change in the state variables.
- It is not necessary to state all the properties (state variables) to define a system completely.

Equation of states

31

- The behaviour of a gaseous system can be described and predicted mathematically as

$$P = f(T, V, n) \quad (1)$$

- The pressure P , is the dependent variable and there are three independent variables, T , V and n . The letter f stands for the functional relationship.
- Eqn (1) phenomenologically summarizes empirical observation called laws or rules i.e. reflect some aspect of the behaviour of nature and must therefore be correct (within limits of experimental error).

Equation of states

32

- The equation of state for one mole of a pure gas is:

$$PV = RT$$

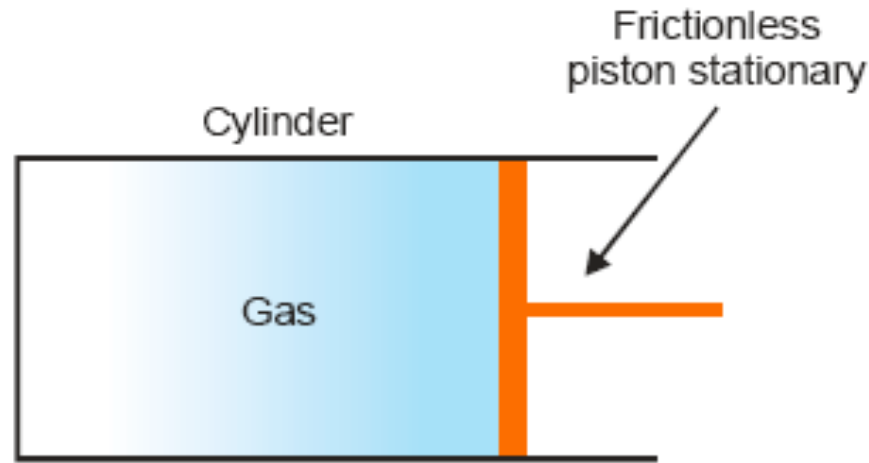
where R is gas constant.

- If of the three state variables (P , V , T), two (P , T) are specified, the value of third (V) becomes fixed and can be calculated from the equation of state.

Equilibrium states

33

- A system in which the state variables have constant values throughout the system is said to be in a state of thermodynamic equilibrium.

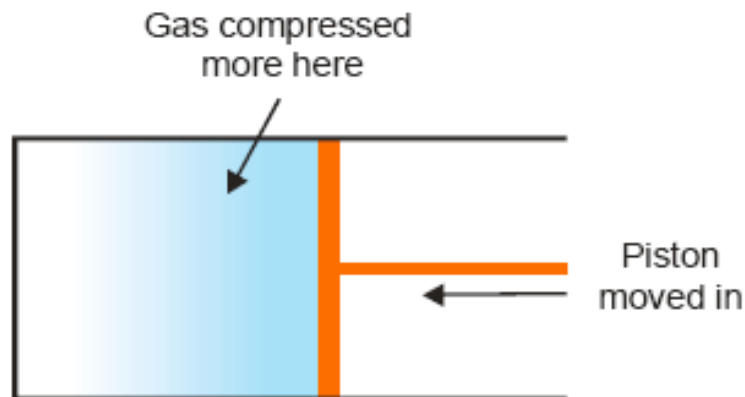


- *If the frictionless piston is stationary, the state of the gas can be specified by giving the values of pressure and volume.*

Non-equilibrium state

34

- A system in which the state variables have different values in different parts of the system is said to be in a non-equilibrium state.



- *If the gas is compressed very rapidly by moving the piston, it passes through states in which pressure and temperature cannot be specified, since these properties vary throughout the gas.*

Criteria for equilibrium

35

- The temperature of the system must be uniform and must be the same as that of the surroundings (**thermal equilibrium**)
- The mechanical properties must be uniform throughout the system (**mechanical equilibrium**)
- The chemical composition of the system must be uniform with no net chemical change (**chemical equilibrium**)

Thermodynamic processes

36

- When a thermodynamic system changes from one state to another, the operation is called a **Process**.
- A process involves change in conditions (temperature, pressure and volume).

Isothermal processes

37

- Those processes in which the temperature remains fixed, are termed isothermal processes.
- Isothermal process is achieved by placing the system in a thermostat (a constant temperature bath).
- For an isothermal process $dT = 0$

Adiabatic processes

38

- Those processes in which no heat flows into or out of the system, are called adiabatic processes.
- Adiabatic conditions can be approached by carrying the process in an insulated container such as 'thermos' bottle. High vacuum and highly polished surfaces help to achieve thermal insulation.
- For an adiabatic process $dq = 0$

Isobaric processes

39

- Those processes which take place at constant pressure are called isobaric processes.
- For example, heating of water to its boiling point and its vaporisation take place at the same atmospheric pressure. These changes are, therefore, designated as isobaric processes and are said to take place isobarically.
- For an isobaric process $dp = 0$

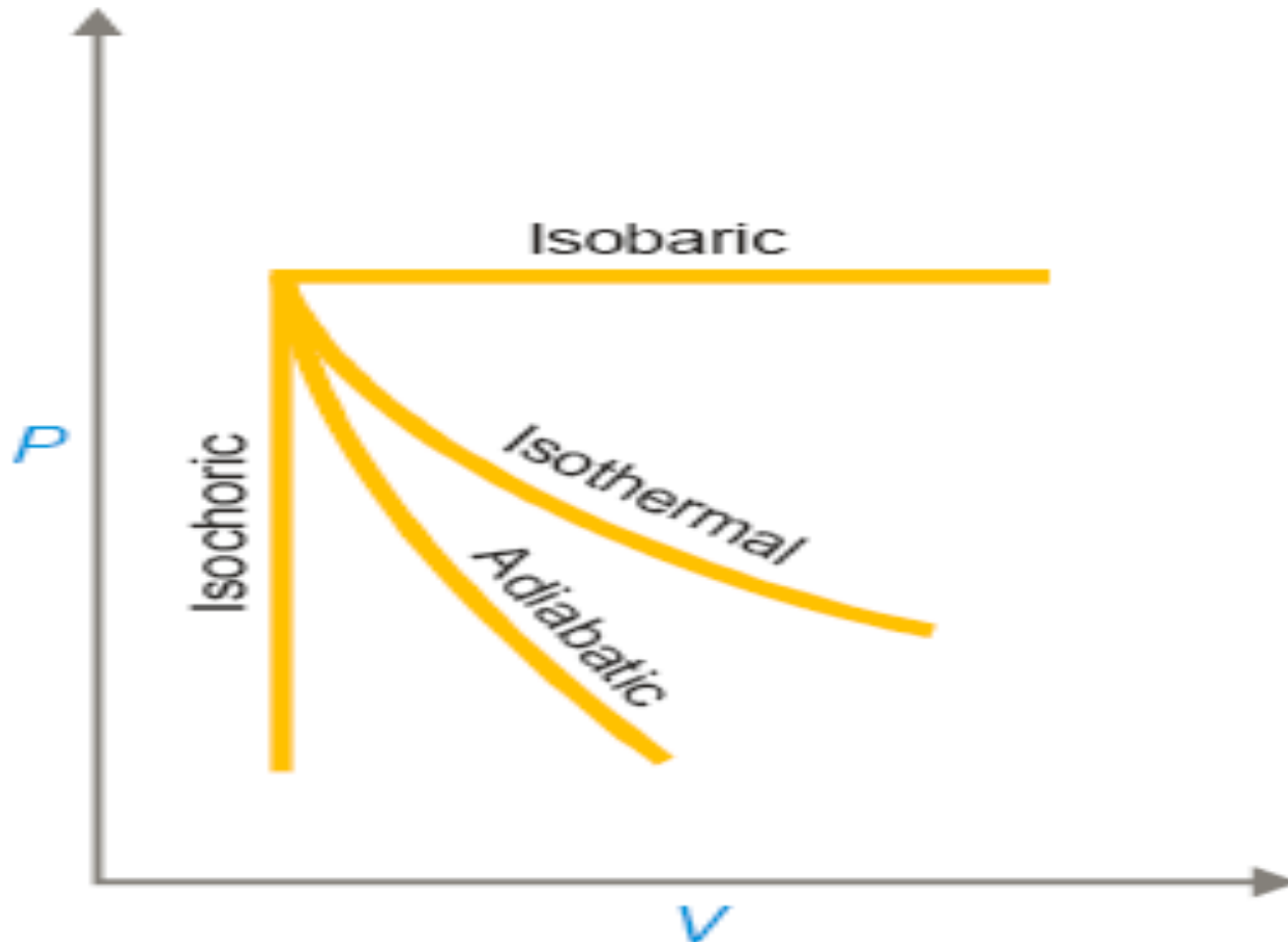
Isochoric processes

40

- Those processes in which the volume remains constant are known as isochoric processes.
- Heating of a substance in a non-expanding chamber is an example of isochoric process.
- For isochoric processes $dV = 0$.

Basic thermodynamic processes

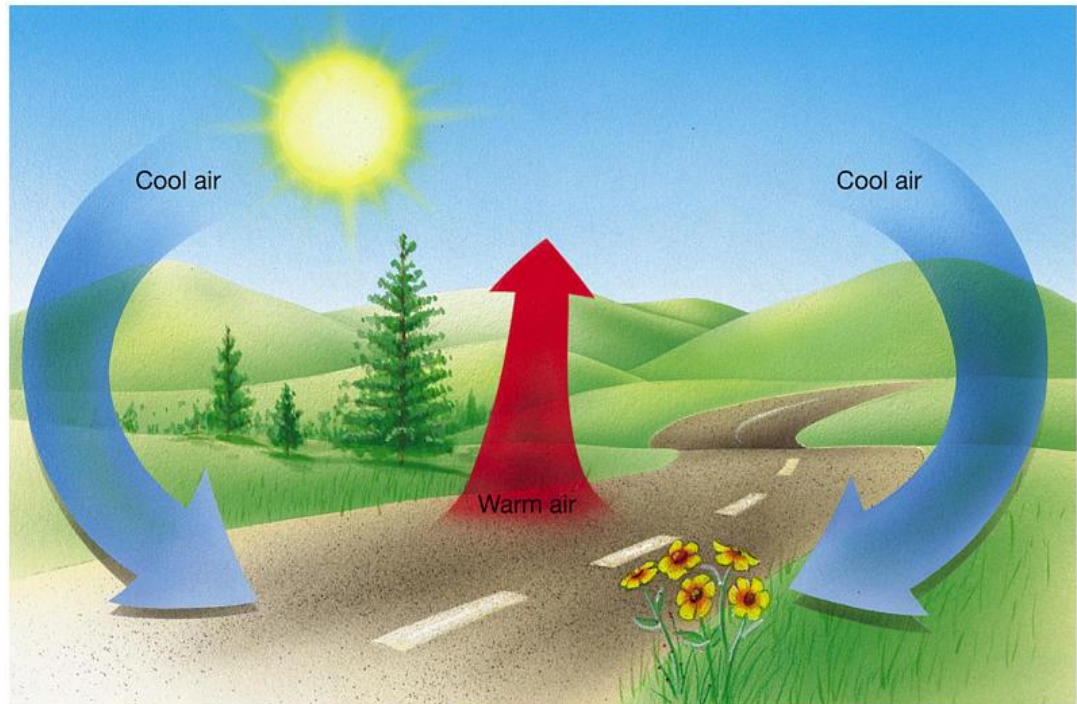
41



Nature of heat and work

42

- **HEAT:** is a transfer of energy as a result of a temperature difference between the system and the surroundings



Work

43

- Work is a transfer of energy that can be used to change the height of a weight somewhere in the surrounding
- Only two ways in which the energy of a closed system can be changed: by transferring energy as work or heat
- We shall refer the term 'work' for mechanical work which is defined as $\text{force} \times \text{distance}$

Units of heat

44

- SI unit of heat is **joule (J)**
- The unit of heat, which was used for many years, is calorie (cal)
- A calorie is defined as the quantity of heat required to raise the temperature of 1 gram of water by 1°C in the vicinity of 15°C

$$1 \text{ joule} = 0.239 \text{ calories}$$

$$1 \text{ calorie} = 4.184 \text{ J}$$

$$1 \text{ kcal} = 4.184 \text{ kJ}$$

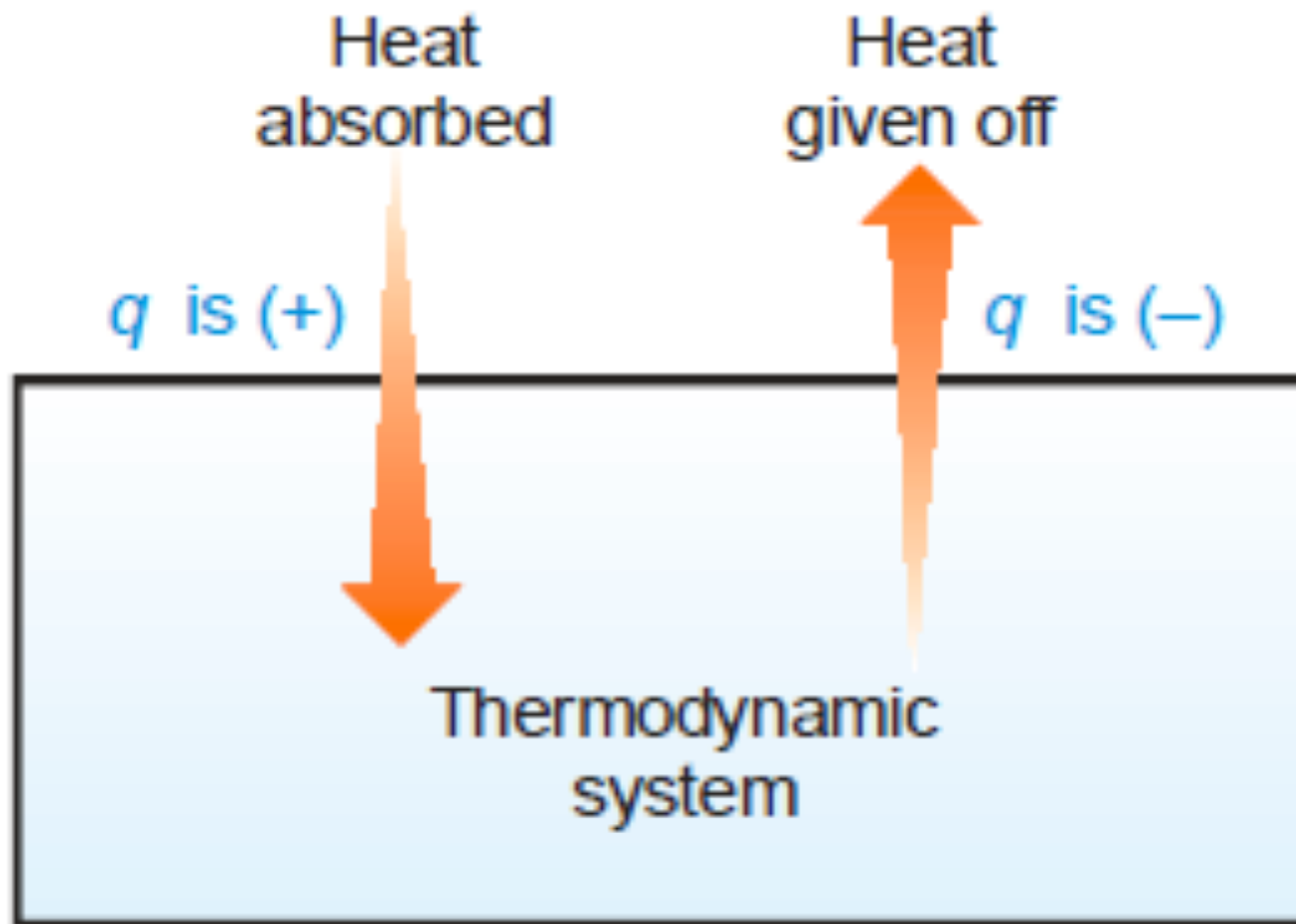
Sign Convention of Heat

45

- The symbol of heat is q
- If the heat flows from the surroundings into the system to raise the energy of the system, it is taken to be positive, $+q$
- If heat flows from the system into the surroundings, lowering the energy of the system, it is taken to be negative, $-q$

Sign convention for heat flow in a system

46



Sign Convention of Work

47

- The symbol of work is w
- If work is done on a system by the surroundings and the energy of the system is thus increased, it is taken to be positive, $+w$
- If work is done by the system on the surroundings and energy of the system is decreased, it is taken to be negative, $-w$

Summary of Sign Conventions

48

Heat flows into the system, q is +ve	Heat flows out of the system, q is -ve
Work is done on the system, w is +ve	Work is done by the system, w is -ve

Pressure–Volume Work

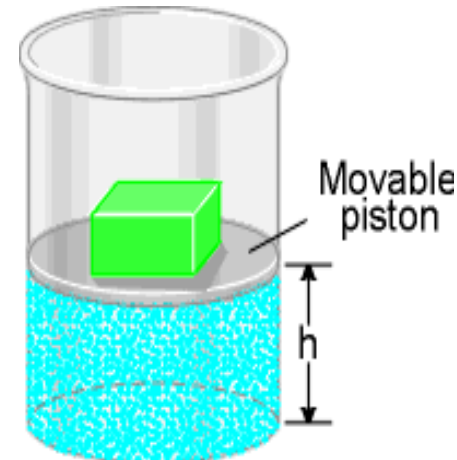
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- Mechanical work is defined as **force multiplied by the distance** through which the force acts
- In elementary thermodynamics, the only type of work generally considered is the work done in expansion (or compression) of a gas
- This is known as pressure-volume work or PV work or expansion work

Pressure–Volume Work

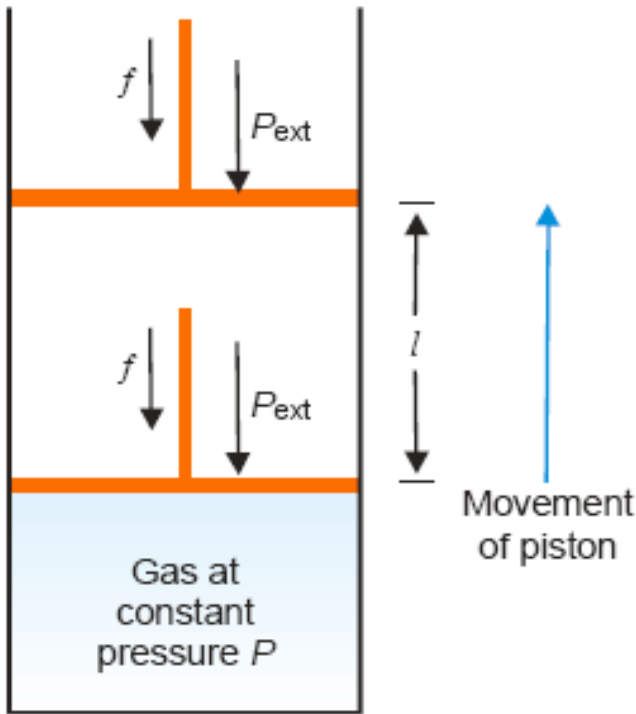
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- For a cylinder fitted with a frictionless piston, the gas exerts a Pressure, P , (force per unit area) on the piston.
- This can be balanced by applying an equal but opposite pressure from outside on the piston (P_{ext}).
- Remember that it is the external pressure (P_{ext}) and not the internal pressure of the gas itself which is used in evaluating work. This is true whether it be expansion or contraction.



Pressure–Volume Work (Expansion)

51



Since the system (gas) is doing work on the surroundings (piston), it bears a –ve sign.

Work = force $\times$ distance

$$w = f \times l$$

(1)

Since pressure is force per area,

$$f = P_{\text{ext}} \times A$$

(2)

Where A is the cross-sectional area of the piston,

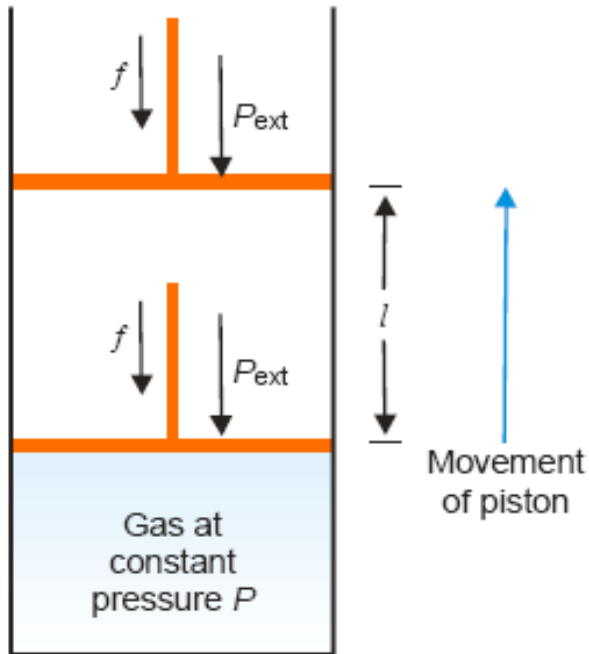
From (1) and (2),

$$w = P_{\text{ext}} \times A \times l$$

$$w = -P_{\text{ext}} \times \Delta V$$

Pressure–Volume Work (Compression)

52



- Similarly, work done in compression of a gas can also be calculated.
- In that case the piston will move down and sign of the work will be positive.

$$\square w = P_{\text{ext}} \times \Delta V$$

Class exercise

53

Calculate the pressure-volume work done when a system containing a gas expands from 1.0 litre to 2.0 litres against a constant external pressure of 10 atmospheres.

Internal energy

54

- A thermodynamic system has within itself a definite quantity of energy.
- This energy includes the translation kinetic, rotational and vibrational energies as well as the kinetic and potential energy of the nuclei and electrons within the individual molecules in the system.
- The total of all the possible kinds of energy of a system, is called its Internal Energy.

Symbol representation of internal energy and sign conventions

55

- The internal energy of a system is represented by the symbol E (Some books use the symbol U).
- It is neither possible nor necessary to calculate the absolute value of internal energy of a system.
- In thermodynamics we are concerned only with the energy changes when a system changes from one state to another.

Symbol representation of internal energy and sign conventions

56

- If ΔE be the difference of energy of the initial state (E_{in}) and the final state (E_f), we can write

$$\Delta E = E_f - E_{in}$$

- ΔE is +ve if E_f is greater than E_{in} and -ve if E_f is less than E_{in} .
- A system may transfer energy to or from the surroundings as heat or as work, or both.

Units of internal energy

57

- The SI unit for internal energy of a system is the joule (J).
- Another unit of energy which is not an SI unit is the calorie,

$$1 \text{ cal} = 4.184 \text{ J.}$$

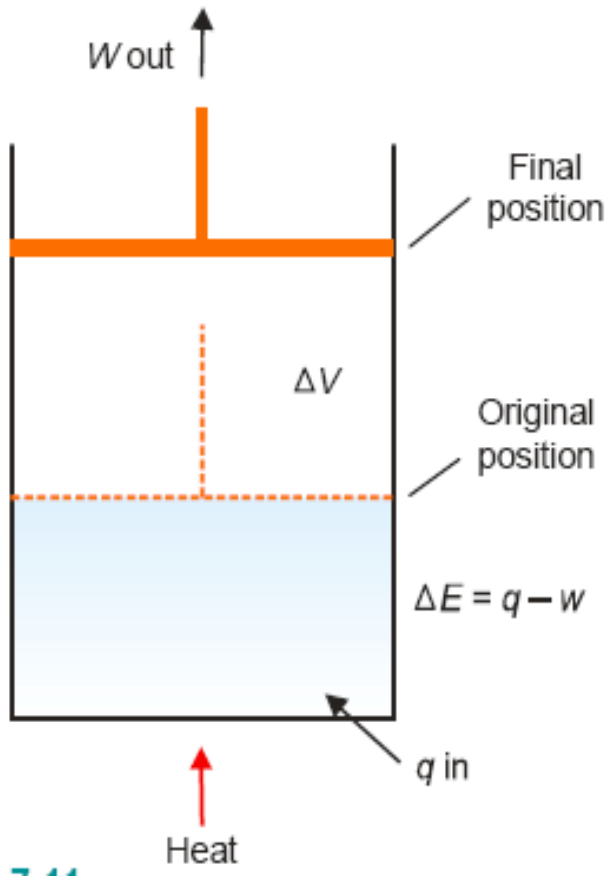
First law of thermodynamics

58

- It states that : the total energy of an isolated system remains constant though it may change from one form to another.
- The first law of thermodynamics is, in fact, an application of the broad principle known as the Law of Conservation of Energy to the thermodynamic system.

First law of thermodynamics

59



- When a system is changed from state A to state B, it undergoes a change in the internal energy from E_A to E_B . Thus,

$$\Delta E = E_B - E_A$$

- This energy change is brought about by the evolution or absorption of heat and/or by work being done by the system.

First law of thermodynamics

60

- Because the total energy of the system must remain constant, we can write the First Law as :

$$\Delta E = q - w$$

where q = the amount of heat supplied to the system

w = work done by the system

- Thus First Law may also be stated as: the net energy change of a closed system is equal to the heat transferred to the system minus the work done by the system.

Other definitions of first law of thermodynamics

61

- 1) Whenever energy of a particular type disappears equivalent amount of another type must be produced.
- 2) Total energy of a system and surroundings remains constant (or conserved)
- 3) It is impossible to construct a perpetual motion machine that can produce work without spending energy on it.

Some special forms of first law of thermodynamics

62

- Mathematical statement of the First law of Thermodynamics is

$$\Delta E = q - w$$

- **Case 1** : For a cyclic process involving isothermal expansion of an ideal gas

$$\Delta E = 0$$

$$\therefore q = w$$

Some special forms of first law of thermodynamics

63

□ **Case 2** : For an isochoric process (no change in volume) there is no work of expansion i.e. $w = 0$. Hence $\Delta E = q_v$

□ **Case 3** : For an adiabatic process there is no change in heat gained or lost i.e. $q = 0$. Hence

$$\Delta E = -w$$

□ i.e., the decrease in internal energy is exactly equal to the work done on the system by surroundings.

Some special forms of first law of thermodynamics

64

- **Case 4** : For an isobaric process there is no change in pressure,

$$\Delta P = 0.$$

- Hence

$$\Delta E = q - w$$

$$\Delta E = q - P\Delta V$$

Enthalpy of a system

65

- In a process carried at constant volume, the heat content of a system is the same as internal energy (E), as no PV work is done.
- But in a constant-pressure process, the system (a gas) uses energy in doing PV work. Therefore, the total heat content of a system at constant pressure is equivalent to the internal energy E plus the PV energy.
- This is called the Enthalpy of the system and is represented by the symbol H.
- Thus enthalpy is defined by the equation :

$$H = E + PV$$

Enthalpy – A function of state

66

- The value of H is dependent on E , P , V which are all state functions.
- Thus, H must also be a function of state.
- Hence the value of H is independent of the path by which the state of the system is changed.

Units and sign conventions of enthalpy

67

- $\Delta H = H_2 - H_1$
- ΔH is positive if $H_2 > H_1$ and the process or reaction will be **Endothermic**.
- ΔH is negative if $H_1 > H_2$ and the reaction will be **exothermic**.



- The quantity of heat 68.32 kcal on the right hand represents $-\Delta H$ of the reaction.

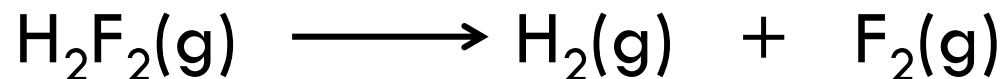
Relation between ΔH and ΔE

68

$$\Delta H = \Delta E + P\Delta V$$

$$\Delta H = \Delta E + \Delta nRT$$

□ For the reaction ,



$$\Delta H = -14.2 \text{ kcal/mole at } 25^\circ \text{ C}$$

Calculate ΔE for the reaction.

Molar heat capacities

69

- **Heat capacity** of a system is the heat absorbed by unit mass in raising the temperature by one degree (K or ° C) at a specified temperature.

$$c = \frac{q}{m \times (T_2 - T_1)}$$

- When mass considered is 1 mole, we can write

$$C = \frac{q}{T_2 - T_1} = \frac{q}{\Delta T}$$

where C is denoted as Molar heat capacity.

Units of heat capacity

71

- The SI unit is joules per degree per mole ($\text{J K}^{-1}\text{mol}^{-1}$).